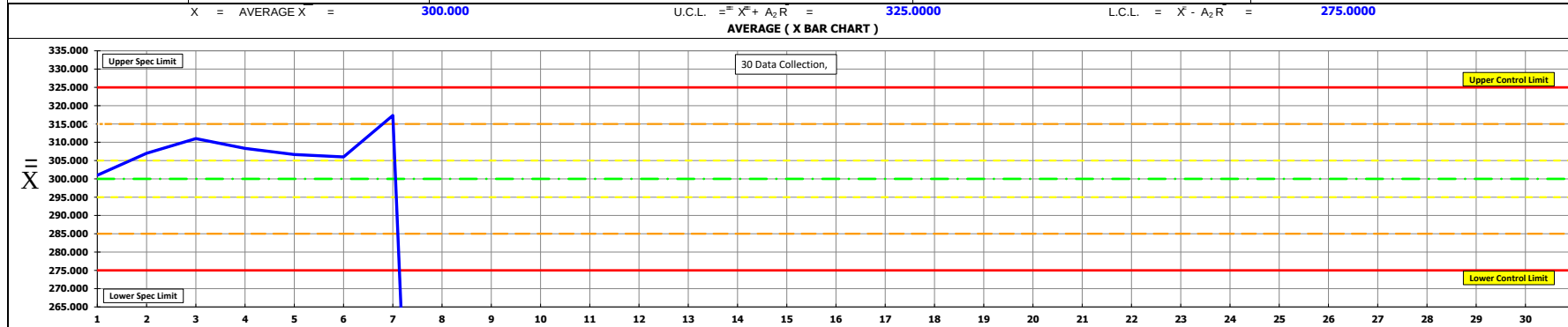


X - R CONTROL CHART

No. Dokumen	4-QAS-PP07-001
Revisi	00
Tgl Berlaku	02-Dec-16

MACHINE CT-1	SECTION QI	OPERATION CT-1	DATE CONTROL LIMITS CALC.		SPECIFICATION min 270 max 330		COURSE NO.	CONTROL ITEM (Δ) YES NO	☐
	MONTH Jun-25	CHARACTERISTIC Temperatur		SAMPLE SIZE / FREQUENCY n = 3		TYPE / SIZE : D2DSW2 / 0.50 PIC = DWI AGUS			

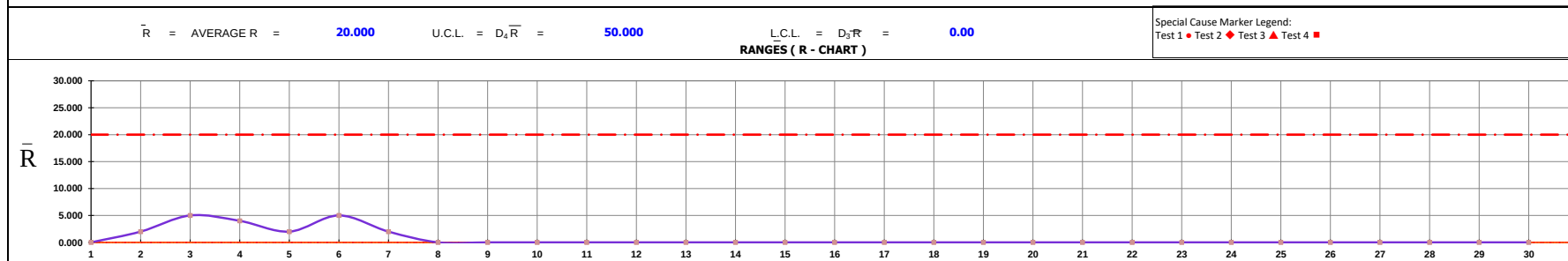


	SPECIAL CAUSES
--	-----------------------

- ✓ TEST 1: ANY POINT OUT OF CONTROL LIMIT
- ✓ TEST 2: 7 POINTS IN A ROW ON SAME SIDE OF CENTER LINE
- ✓ TEST 3: 6 POINTS IN A ROW ALL INCREASING OR DECREASING
- ✓ TEST 4: 14 POINTS IN A ROW, ALTERNATING UP & DOWN
- ✓ TEST 5: 2 OUT OF 3 POINTS > 2 STDEV FROM CENTER LINE (SAME SIDE)
- ✓ TEST 6: 4 OUT OF 5 POINTS > 1 STDEV FROM CENTER LINE (SAME SIDE)
- ✓ TEST 7: 15 POINTS IN A ROW WITHIN 1 STDEV OF CENTER LINE (EITHER SIDE)
- ✓ TEST 8: 8 POINTS IN A ROW > 1 STDEV FROM CENTER LINE (EITHER SIDE)

ACTION INSTRUCTION

1. _____
2. _____
3. _____
4. _____
5. _____



SUBGROUP SIZE	A ₂	d ₂	D ₃	D ₄
2	1.881	1.128	0	3.267
3	1.023	1.693	0	2.574
4	0.729	2.059	0	2.281
5	0.577	2.326	0	2.115
6	0.483	2.534	0	2.004
7	0.4193	2.704	0.0756	1.9244
8	0.3726	2.847	0.1361	1.8639

[illegible]

CALCULATION

	X =	308.2		
	R =	3.3		
	S _{ST} =	2.0	S _{LT} =	4.996

THE PROCESS MUST BE IN CONTROL

BEFORE CAPABILITY CAN BE DETERMINED

$$C_{PK} = \min \left[\frac{X - LSL}{3s_{ST}}, \frac{USL - X}{3s_{ST}} \right]$$

$$= \min [5.62, 2.85]$$

$$= 2.85$$

$$C_p = \left[\frac{USL - LSL}{6\sigma_T} \right]$$

$$= \left[\frac{4.23}{6} \right]$$

CHECKED BY

APPROVED BY _____

[illegible]